

Computer Organization & Architecture

Unit- 1(Part 1) Structure of Computers

Prepared By: Sweetu D. Sureja
Information Technology Department
V.V.P. Engineering College, Rajkot.

Index

- Computer types
- Functional units
- Von Neumann Architecture
- Basic operational concepts
- Bus Structures
- Software
- Multiprocessors and Multicomputer
- Data representation
- Fixed and Floating point
- Error detection and correction codes.

Computer types

Classification of Computers



BY TYPE

ANALOG

DIGITAL

HYBRID

BY SIZE

MICRO

MINI

MAINFRAME

SUPER

BY PURPOSE

GENERAL PURPOSE

SPECIAL PURPOSE

By types – Analog, Digital and Hybrid

Type of Computer	Description / Features	Uses / Examples
Analog Computer	Used to solve complex scientific and industrial problems. Provides quick results where exact values are not required. Has limited accuracy and complex structure, so it is not commonly used for general purposes today.	Scientific and industrial applications
Digital Computer	Solves problems in discrete form using binary data (0s and 1s). Can store, process, and manage large amounts of data efficiently. Provides accurate and reliable control of operations.	General-purpose computing, data processing
Hybrid Computer	Combines features of both analog and digital computers. Logical operations are handled by digital components, while differential equations are solved using analog components.	Space flights, food processing plants

By size -

Feature	Micro Computer	Mini Computer	Mainframe Computer	Super Computer
Size	Very small	Medium	Very large	Extremely large
Cost	Low	Medium	High	Very high
Processing Power	Low to moderate	Moderate	High(million ins.)	Very high(billion ins.)
Users	Single user	Multiple users	Hundreds to thousands	Used by experts
Data Handling	Small amount	Medium amount	Very large amount	Massive data
Applications	Personal and office work	Control systems, small servers	Scientific and business applications, Banking, trading	Weather forecasting, research, Defense research, Aircrafts design
Examples	Desktop, Laptop	Smartphones, iPads	Bank systems	Climate models, space research

By Purpose

- **General Purpose Computers**

- Designed to perform **multiple tasks**.
- Can handle **input/output operations and calculations**.
- Can store **small to moderate data**.
- Used for **general activities** like typing, browsing, and communication.
- Examples: **Calculator, Desktop, Laptop, Mobile phone**.
-

- **Special Purpose Computers**

- Designed to perform **one specific task**.
- Used where **high accuracy and efficiency** are required.
- Customized for **particular applications**.
- Examples: **Thermometer, Generator, Climate analysis system, Manufacturing machines**.

Von Neumann Architecture

- In 1945 John von Neumann was the first published architecture of the computer, which was known as Von Neumann architecture.
- According to Von Neumann Architecture, The basic function performed by a computer is the execution of a program.
- A program is a set of machine instructions.
- A typical Von Neumann machine consists of five functionally independent units :
 - 1) Input unit
 - 2) Output unit
 - 3) Arithmetic and logic units
 - 4) Control unit
 - 5) Memory unit

Functional units

(1) Input Unit

- Accepts data and instructions from the user
- Examples: Keyboard, Mouse, Scanner

(2) Output Unit

- Displays or produces results
- Examples: Monitor, Printer, Speaker

(3) Arithmetic Logic Unit (ALU)

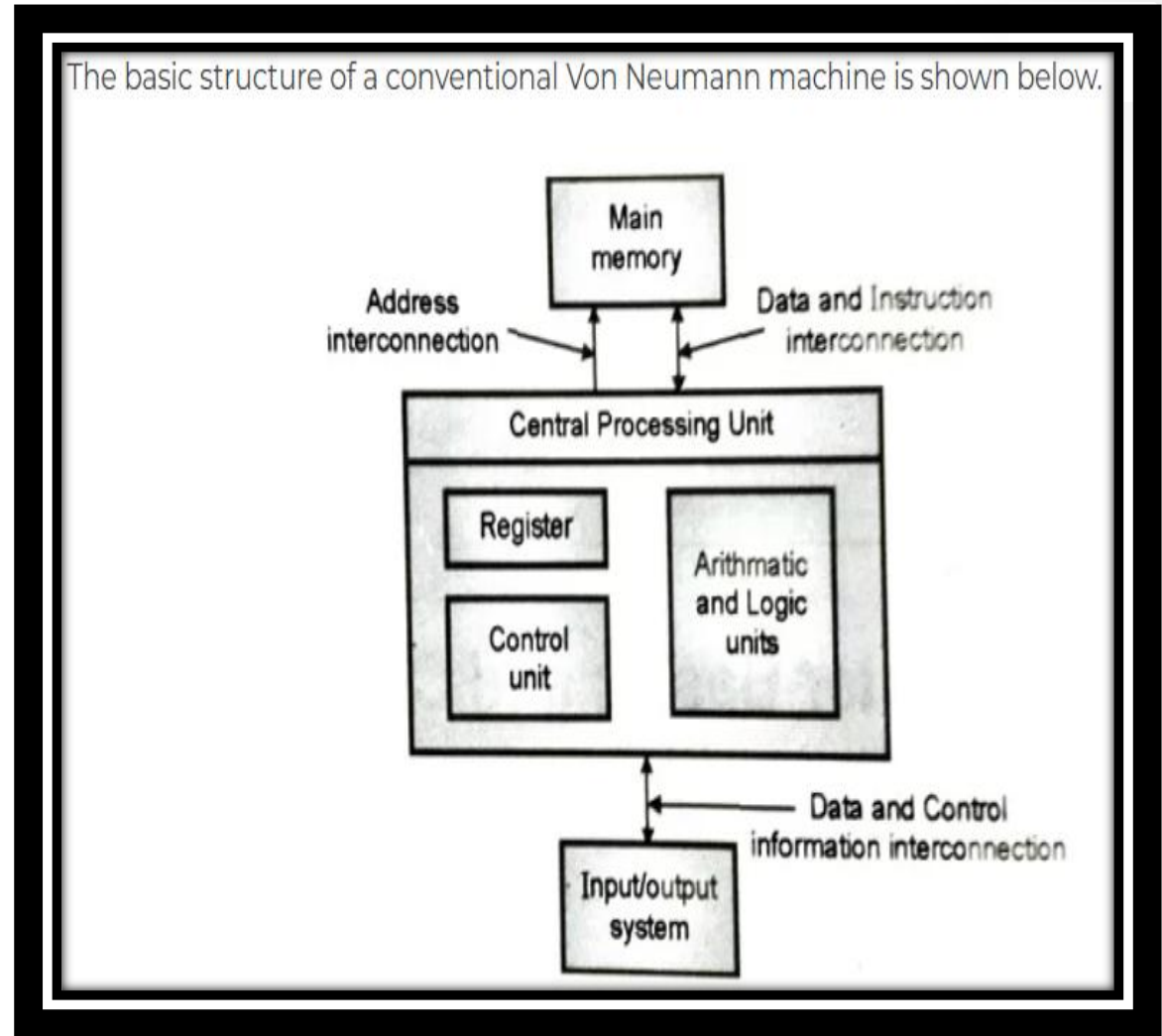
- Performs arithmetic operations (+, −, ×, ÷)
- Performs logical operations (AND, OR, NOT)

(4) Control Unit (CU)

- Directs all operations of the computer
- Controls data flow between memory, ALU, and I/O devices

(5) Memory Unit

- Stores data and instructions.
- **Primary Memory:** RAM, ROM
- **Secondary Memory:** Hard disk, SSD, Pen drive



Basic operational concepts

- The basic operation of a computer follows these steps:
- **Fetch** – Instruction is fetched from memory
- **Decode** – Instruction is decoded by control unit
- **Execute** – ALU performs required operation
- **Store** – Result is stored back in memory

- Computers fundamentally operate by **Input, Processing, Storage, Output, and Control**, fetching instructions from memory, executing them in the CPU (using ALU and registers like PC, IR, MAR, MDR), and moving data between memory and CPU to perform tasks, ultimately delivering results.
- This cycle, known as the **Instruction Cycle**, involves fetching, decoding, executing instructions, and storing results.

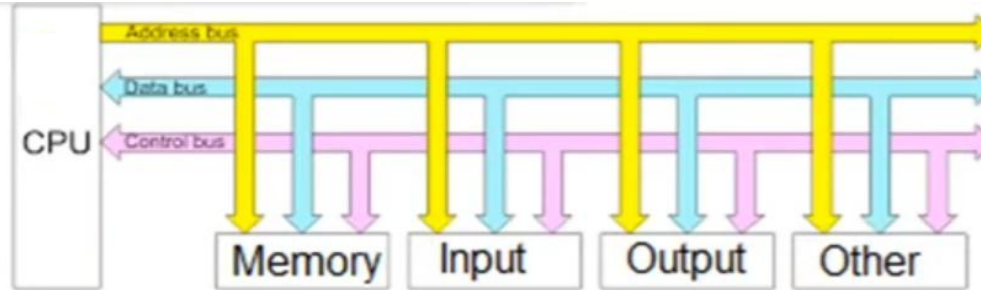
Example

Add two numbers:

1. Input two numbers
2. Store numbers in memory
3. Fetch addition instruction
4. Decode instruction
5. Execute addition in ALU
6. Store result
7. Display output

Bus Structures

Types of Buses:

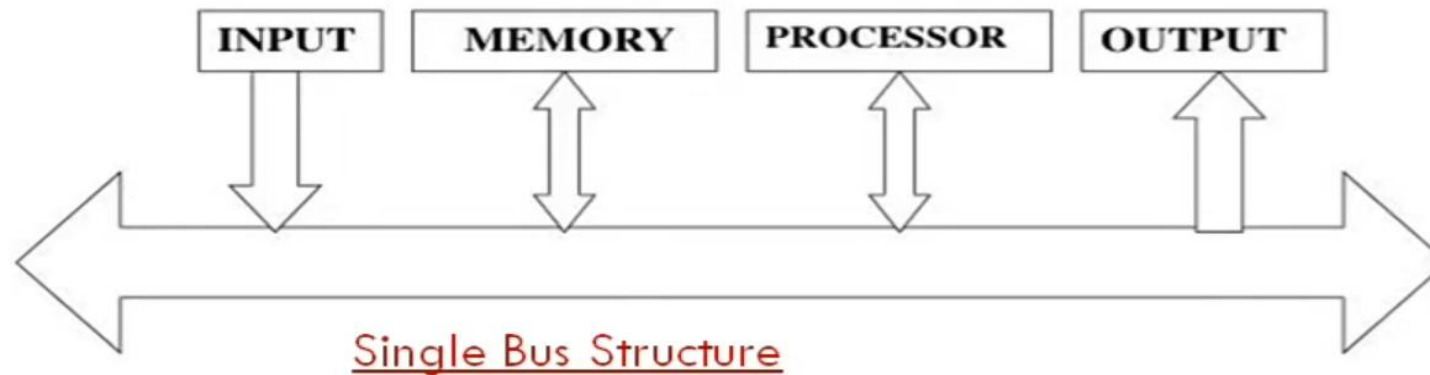


- ❑ **Address bus: unidirectional** : group of wires which carries *address information* bits from processor to peripherals
 - ❑ Address bus width determines the maximum memory capacity
 - ❑ For eg: If address line = 3 bits $2^3 = 8$ i.e. 3 address line is required to select 8 location
 - ❑ In general $2^x = n$ where x number of address lines (address bit) and n is number of location
- ❑ **Data bus: bidirectional** : group of wires which carries *data information* bit from processor to peripherals and vice – versa
 - ❑ Move data/instruction between CPU and peripherals
 - ❑ Data bus width determines the system performance (word length of computer)
- ❑ **Control bus: bidirectional:** group of wires which carries *control signals* from processor to peripherals and vice – versa
 - ❑ Control the access to and the use of address lines and data lines
 - ❑ Control signals like memory read/write, I/O read/write, etc

Bus Structures

Single-Bus Structure

- The simplest way to interconnect functional units is to use the **single bus**.
- **Single bus structure:** Common bus used to communicate between peripherals and microprocessor



- **Single bus does one transfer at a time, so only 2 units can actively use the bus** at a given time. *Bus control lines* are used to arbitrate multiple requests for use of one bus.

Bus Structures

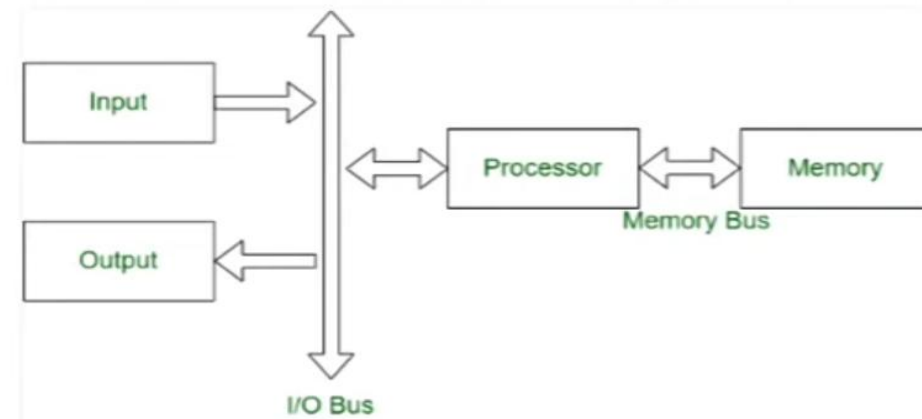
Single-Bus Structure...

- ❑ Main Advantage
 - ▣ Simple and Low cost
 - ▣ Very flexible for attaching peripheral devices
- ❑ Drawback:
 - ▣ Speed Issue: Devices connected to the bus vary widely in their speed of operation.
 - If the speed of bus is bounded by the slowest device connected to it, the efficiency will be very low.
 - ▣ Efficient transfer mechanism is needed to cope with this problem
 - A common approach is to include **buffer registers** with the devices to hold the information during transfers
 - An another approach is to use **Multiple Bus structure** (or two-bus structure) and an additional transfer mechanism

Bus Structures

Multi-Bus Structure

- ❑ System that contain **multiple buses** achieve more parallelism. This leads to better performance but an increase cost.
- ❑ **Multi-bus structure** improves the **performance**, but also **increases cost** significantly
- ❑ In **multi-bus structure**, each of which connects subset of modules
- ❑ Example: In two-bus structure : One bus can be used to fetch instruction other can be used to fetch data, required for execution.



Two-bus structure

Software

- Software is a set of programs and instructions that tell the computer how to perform tasks.
- It acts as an interface between the **user and the hardware**.
- Without software, hardware cannot function.

1. System Software

- Starts automatically when the computer is turned on.
- System software is designed to **control, manage, and operate computer hardware**.
- It provides a **platform for application software**.

Functions

- Controls hardware components such as CPU, memory, and I/O devices.
- Performs memory management and allocation.
- Provides security and error handling.
- Enables communication between hardware and software.

Software

Types of System Software

- **Operating System**
 - Manages overall computer operations.
 - Examples: Windows, Linux, macOS.
- **Device Drivers**
 - Allow OS to communicate with hardware devices.
 - Examples: Printer driver, keyboard driver, cd/dvd driver.
- **Utility Software**
 - Performs system maintenance tasks.
 - Examples: Antivirus, disk cleanup.
- **Language Translators**
 - Convert programs into machine language.
 - Examples: Compiler, Interpreter, Assembler.

Software

2. Application Software

- Application software is designed to **perform specific tasks for users**.
- Cannot run without system software.
- Task-oriented.

Functions

- Helps users perform day-to-day activities.
- Solves user-specific problems.
- Improves productivity and efficiency.
- Provides user-friendly interfaces.

Software

Types of Application Software

- **General-Purpose Software**
 - Used for common tasks.
 - Examples: MS Word, Excel, PowerPoint.
- **Special-Purpose Software**
 - Designed for a specific task.
 - Examples: Payroll system, railway reservation.
- **Customized Software**
 - Developed for specific organizational needs.
 - Examples: Banking software, college management system.

Multiprocessor and Multicomputer

Parameter	Multiprocessor System	Multicomputer System
Definition	A system having multiple processors within a single computer	A system having multiple independent computers connected by a network
Processors	Two or more CPUs in one system	Each computer has its own CPU
Memory	Shared main memory among processors	Separate memory for each computer
Operating System	Single operating system controls all processors	Multiple operating systems , one per computer
Communication Method	Through shared memory	Through message passing
Communication Speed	Very fast	Relatively slower
Scalability	Limited scalability	Highly scalable
Cost	High cost hardware	Lower cost (uses normal computers)
Reliability	Failure of one processor reduces performance	Failure of one node does not stop the system
Programming Complexity	Less complex	More complex
Examples	Multicore CPUs, servers	Clusters, cloud systems

Data representation

Basic computer data types & Complements

Basic computer data types

- Binary information is stored in **memory** or **processor registers**.
- Registers either contain **data** or **control information**.
- The data types(given by user through keyboard) found in registers of digital computers may be classified as follows:
 1. Numbers used in arithmetic computations (Eg.0.....9)
 2. Letters of the alphabet used in data preprocessing (Eg.A.....Z,a...z)
 3. Other symbols used for specific purpose (Eg. @, #)
- The binary number system is the most natural system to use in a digital computer.
- **Why Binary number system.....???????**
 - Registers are made up of flip-flops and flip-flops are two state devices that can only store 1's and 0's.

Number System

Type	Base(Radix)	Number of digits
Decimal number system	10	0-9
Binary number system	2	0 & 1
Octal number system	8	0-7
Hexadecimal number system	16	0-9, A-F

Decimal Numbers	Binary Numbers	Octal Numbers	Hexa-decimal Numbers
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	-	8
9	1001	-	9
10	1010	-	A
11	1011	-	B
12	1100	-	C
13	1101	-	D
14	1110	-	E
15	1111	-	F

Conversion of Number Systems

1. Conversion from Decimal Number System to Other Number Systems
2. Conversion from Binary Number System to Other Number Systems
3. Conversion from Octal Number System to Other Number Systems
4. Conversion from Hexadecimal Number System to Other Number Systems

1. Conversion from Decimal Number System to Other Number Systems

A. Decimal to Binary Conversion:

$(243)_{10} \longrightarrow (?)_2$

2	243	1
2	121	1
2	60	0
2	30	0
2	15	1
2	7	1
2	3	1
	1	

$\longrightarrow (11110011)_2$

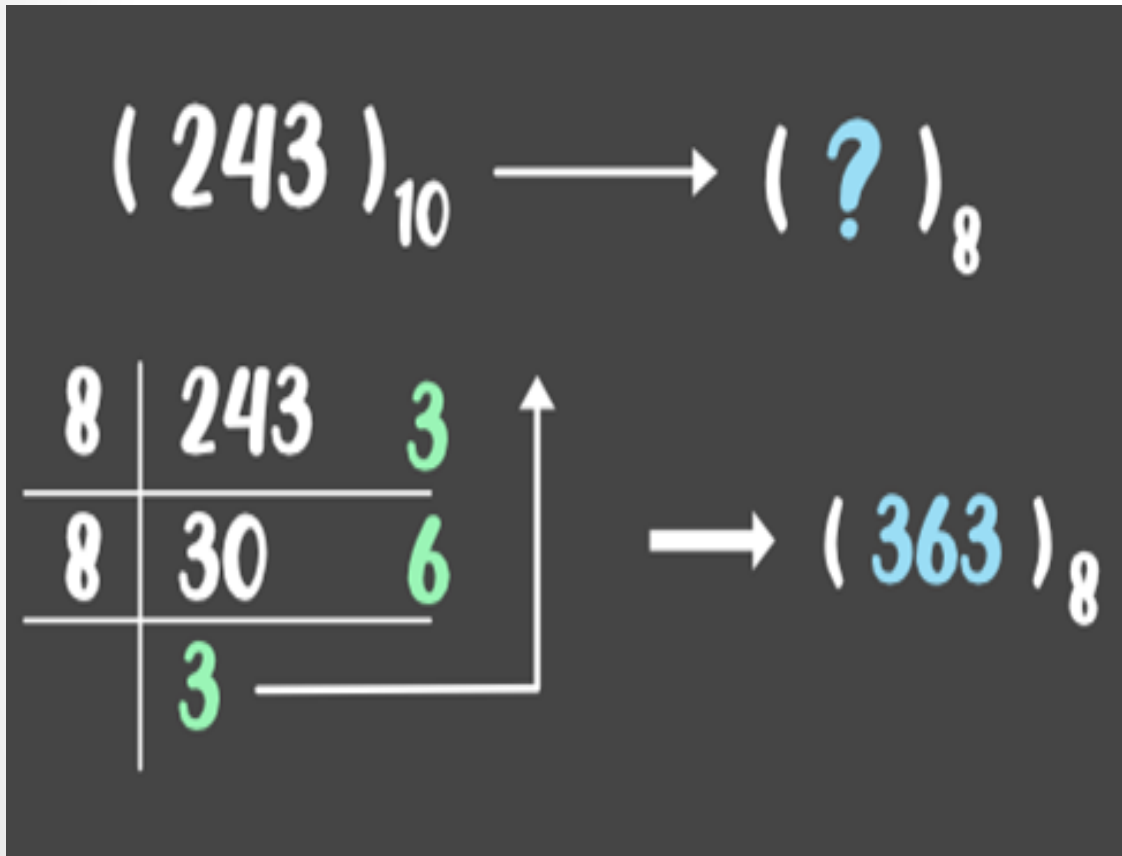
1. Conversion from Decimal Number System to Other Number Systems

B. Decimal to Octal Conversion:

$(243)_{10} \longrightarrow (?)_8$

8	243	3
8	30	6
	3	

$\longrightarrow (363)_8$



1. Conversion from Decimal Number System to Other Number Systems

C. Decimal to Hexadecimal Conversion:

$(243)_{10} \longrightarrow (?)_{16}$

16	243	3
	15	

$\longrightarrow (15\ 3)_{16} \longrightarrow (F3)_{16}$

2. Conversion from Binary Number System to Other Number Systems

A. Binary to Decimal Conversion:

$$(11101011)_2 \longrightarrow (?)_{10}$$

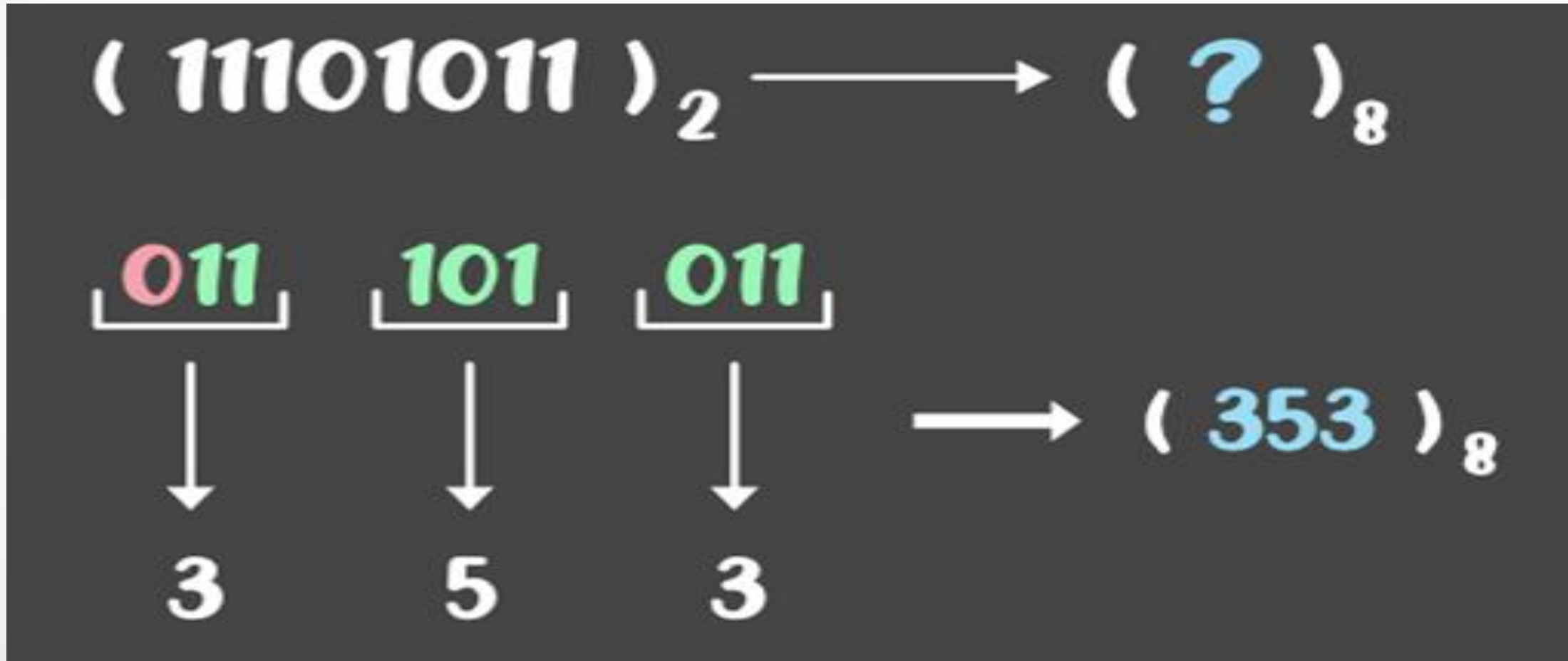
$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$128 + 64 + 32 + 0 + 8 + 0 + 2 + 1$$

$$(235)_{10}$$

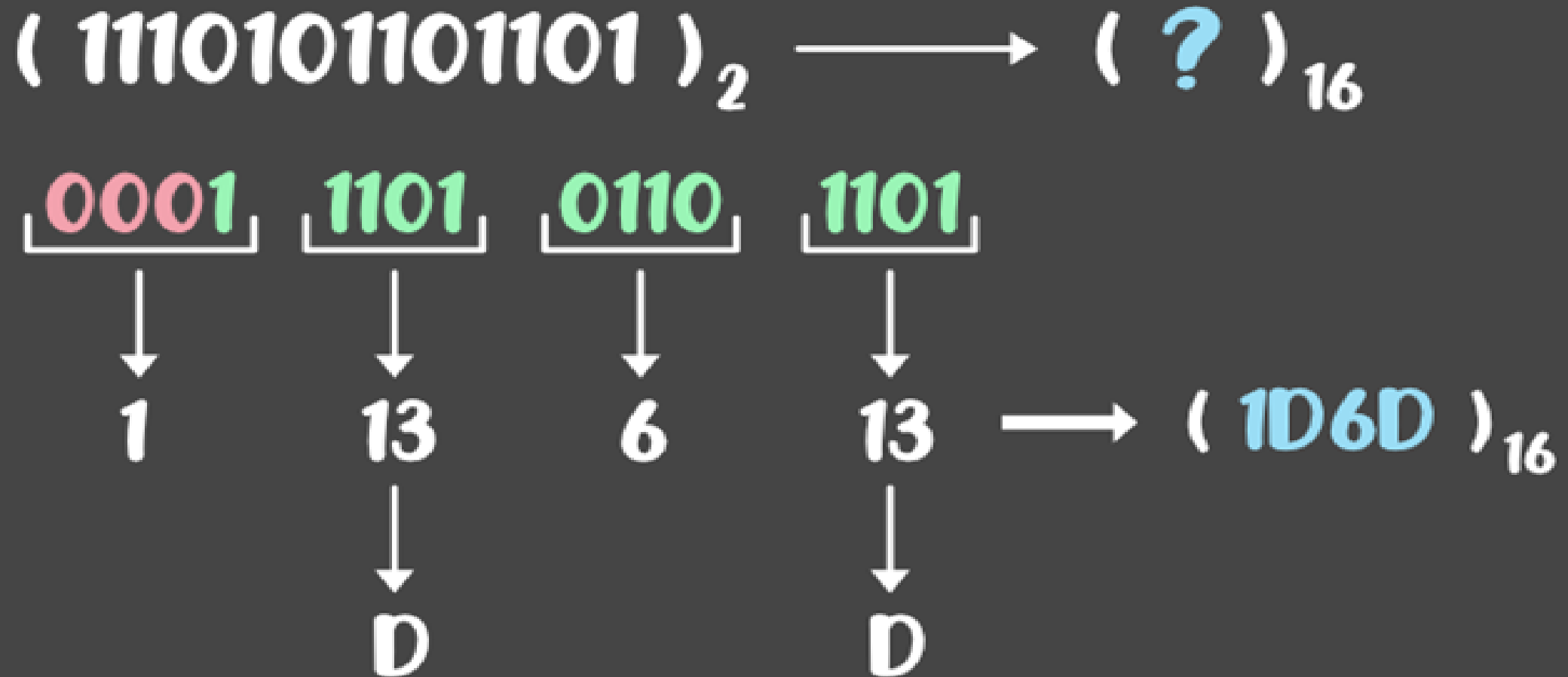
2. Conversion from Binary Number System to Other Number Systems

B. Binary to Octal Conversion:



2. Conversion from Binary Number System to Other Number Systems

C. Binary to Hexadecimal Conversion:



3. Conversion from Octal Number System to Other Number Systems

A. Octal to Decimal Conversion:

$$(247)_8 \longrightarrow (?)_{10}$$

$$2 \times 8^2 + 4 \times 8^1 + 7 \times 8^0$$

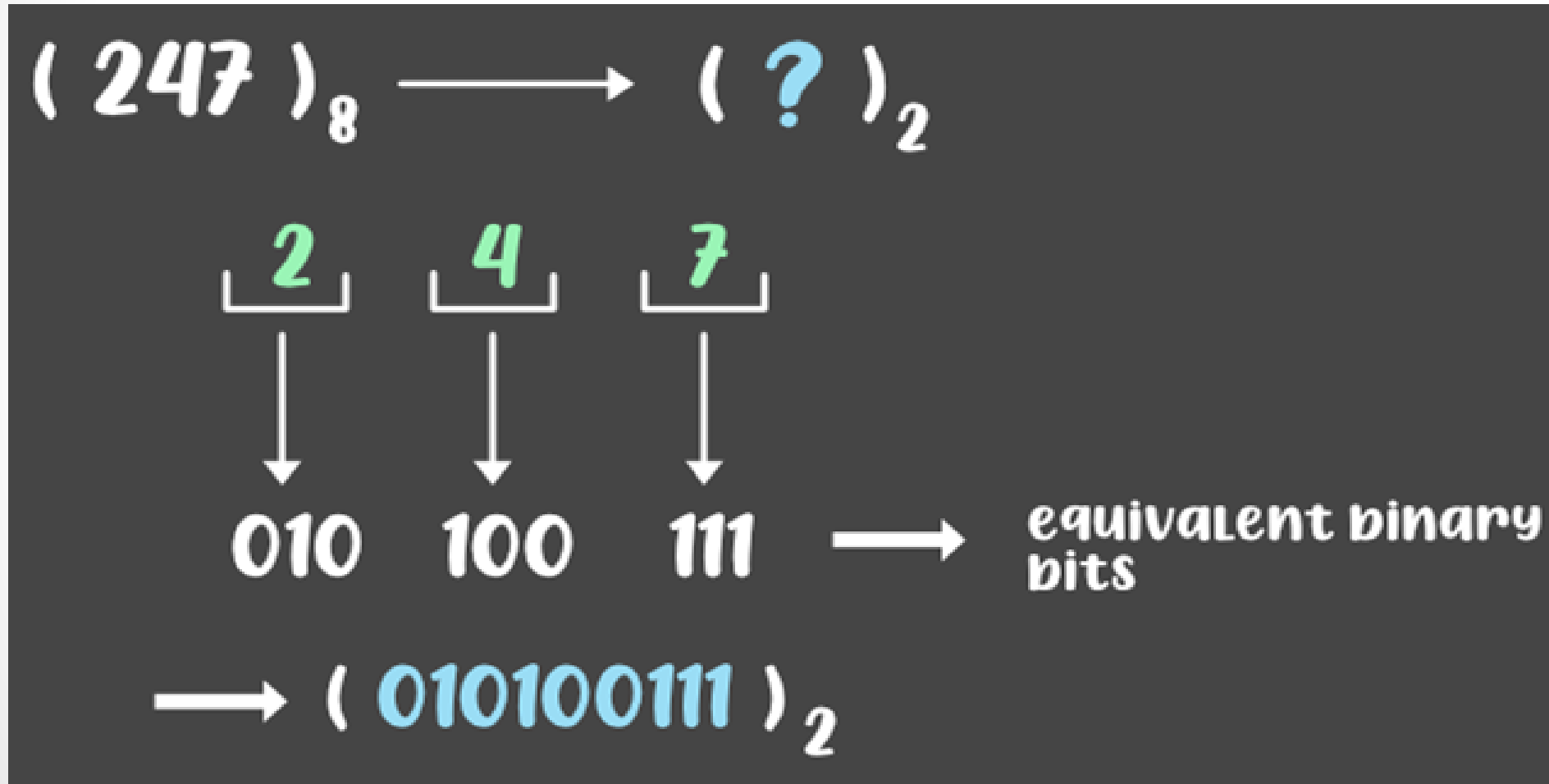
$$2 \times 64 + 4 \times 8 + 7$$

$$128 + 32 + 7$$

$$(167)_{10}$$

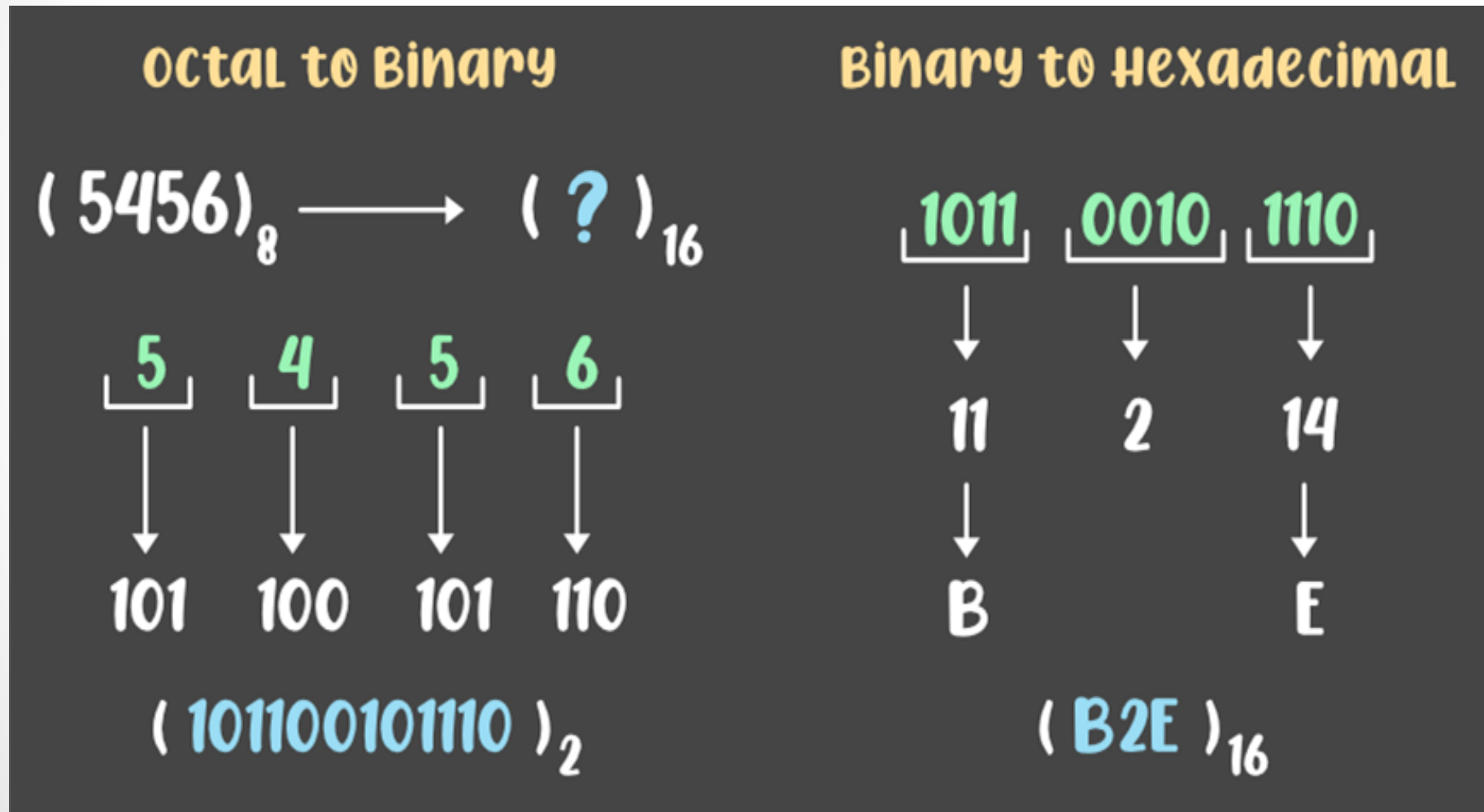
3. Conversion from Octal Number System to Other Number Systems

B. Octal to Binary Conversion:



3. Conversion from Octal Number System to Other Number Systems

C. Octal to Hexadecimal Conversion:



4. Conversion from Hexadecimal Number System to Other Number Systems

A. Hexadecimal to Decimal Conversion:

$$(8EB4)_{16} \longrightarrow (?)_{10}$$

8 14 11 4

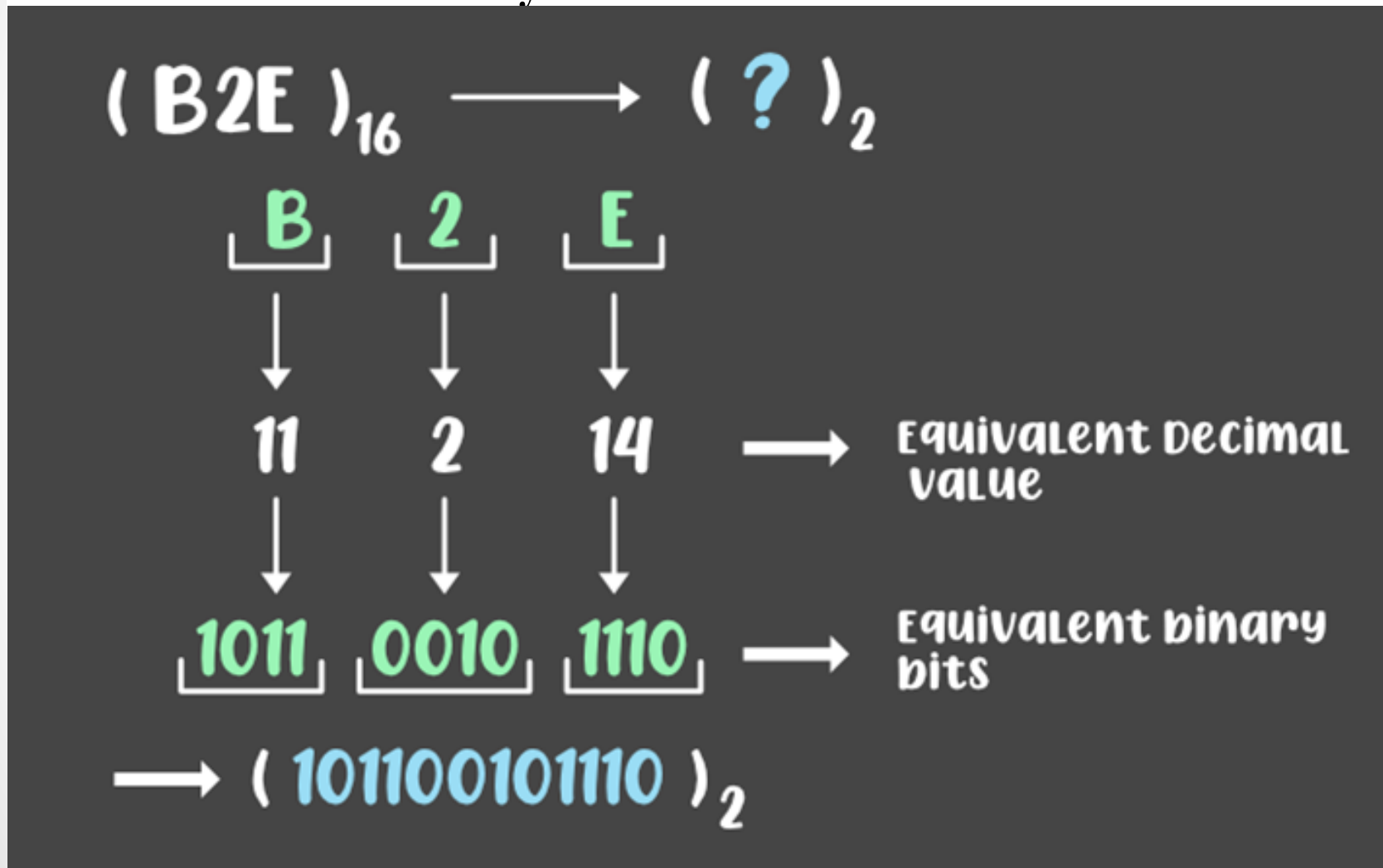
$$8 \times 16^3 + 14 \times 16^2 + 11 \times 16^1 + 4 \times 16^0$$

$$32768 + 3584 + 176 + 4$$

$$(36532)_{10}$$

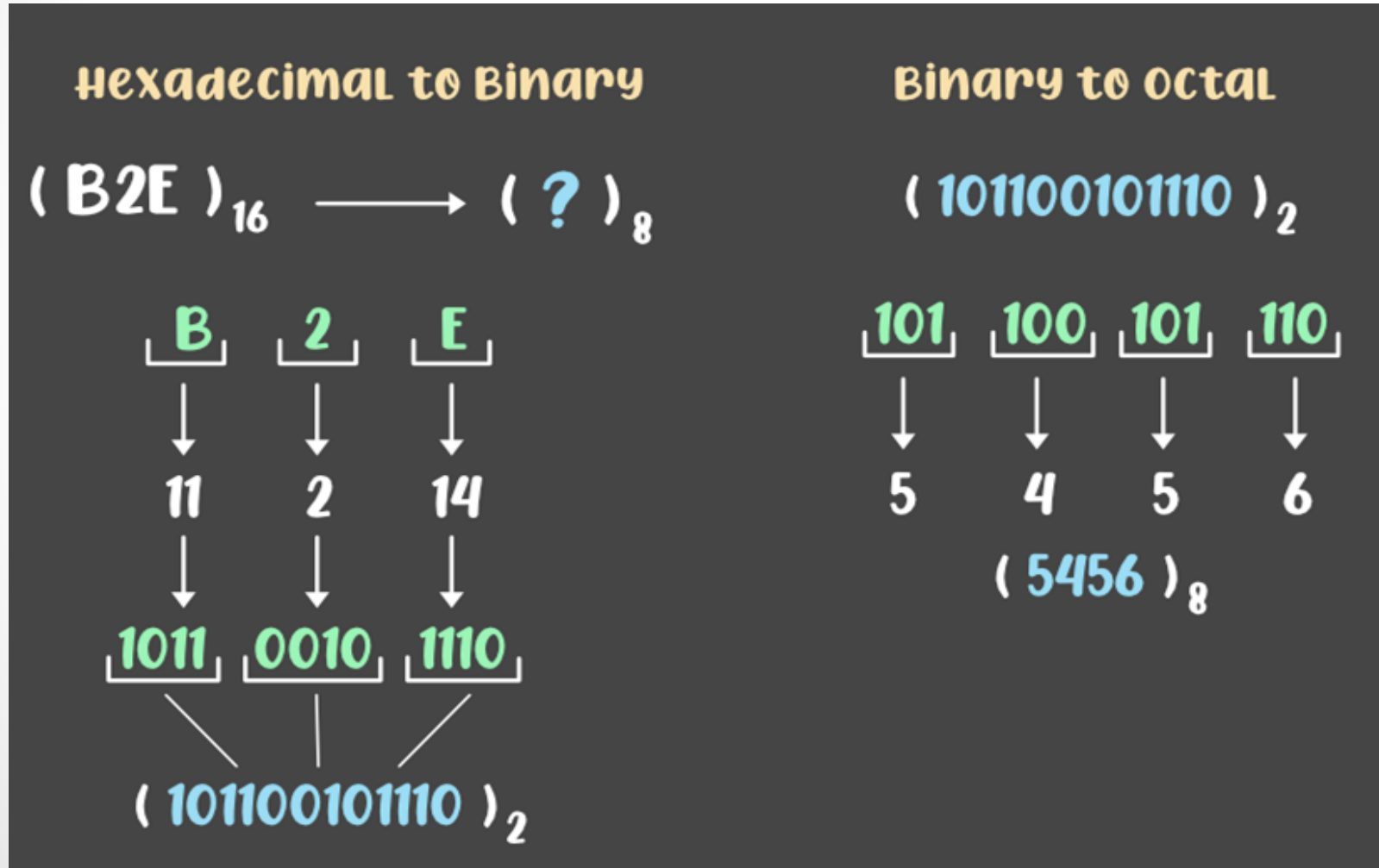
4. Conversion from Hexadecimal Number System to Other Number Systems

B. Hexadecimal to Binary Conversion:



4. Conversion from Hexadecimal Number System to Other Number Systems

C. Hexadecimal to Octal Conversion:



Conversion of Number Systems(Examples)

- $(304)_8 = (\quad ? \quad)_{10}$
- $(1534)_8 = (\quad ? \quad)_{10}$
- $(A10)_{16} = (\quad ? \quad)_{10}$
- $(BCA)_{16} = (\quad ? \quad)_{10}$
- $(11011001)_2 = (\quad ? \quad)_8$
- $(10110011)_2 = (\quad ? \quad)_8$
- $(1111010100)_2 = (\quad ? \quad)_{16}$
- $(1111010100111110)_2 = (\quad ? \quad)_{16}$
- $(425)_{10} = (\quad ? \quad)_8$
- $(1423)_{10} = (\quad ? \quad)_{16}$

Conversion of Number Systems(Examples)

- $(304)_8 = (196)_{10}$
- $(1534)_8 = (860)_{10}$
- $(A10)_{16} = (2576)_{10}$
- $(BCA)_{16} = (3018)_{10}$
- $(11011001)_2 = (331)_8$
- $(10110011)_2 = (263)_8$
- $(1111010100)_2 = (3D4)_{16}$
- $(1111010100111110)_2 = (F53E)_{16}$
- $(425)_{10} = (651)_8$
- $(1423)_{10} = (58F)_{16}$